$|\psi\rangle = \begin{pmatrix}1\\0\end{pmatrix} \rightarrow \langle\psi| = (1 \;\; 0)$

$|\phi\rangle = \begin{pmatrix}0\\1\end{pmatrix} \rightarrow \langle\phi| = (0 \;\; 1)$

INNER PRODUCT

$\langle\psi|\phi\rangle \longrightarrow$ NUMBER

$(1 \;\; 0)\begin{pmatrix}0\\1\end{pmatrix} = 0$

OUTER PRODUCT

$|\psi\rangle\langle\phi| \longrightarrow$ MATRIX

$\begin{pmatrix}1\\0\end{pmatrix}(0 \;\; 1) = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix}$

WRITE A MATRIX

$$U = \begin{array}{c c} & \begin{array}{cc}\langle 0| & \langle 1|\end{array} \\ & \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}\end{pmatrix} \begin{array}{c}|0\rangle \\ |1\rangle\end{array}\end{array}$$

$\Rightarrow a_{11} = \langle 0|U|0\rangle$

$|z_+\rangle = \begin{bmatrix}1\\0\end{bmatrix}$ $\quad |x_+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$ $\quad |y_+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\i\end{bmatrix}$

$|z_-\rangle = \begin{bmatrix}0\\1\end{bmatrix}$ $\quad |x_-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$ $\quad |y_-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-i\end{bmatrix}$

$\sum_n |n\rangle\langle n| = \mathbb{1}$

$nn^* = |n|^2$

$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^*$

UNCERTAINTY RELATION $\Delta A \Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$

SCHRODINGER EQUATION $i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$

- NORMALIZE: $\langle\psi|\psi\rangle = 1$
- ORTHONORMAL: $\langle\phi|\psi\rangle = 0$
- NORMAL: $NN^\dagger = N^\dagger N$
- HERMITIAN/ADJOINT: $A = A^\dagger$
- UNITARY: $UU^\dagger = \mathbb{1}$
- TWO OPERATORS COMMUTE IF: $[A,B] = 0$
- TWO STATES ARE ORTHOGONAL IF: $\langle\omega_1|\omega_2\rangle = 0$

• PAULI MATRICES

"INVERSE of IDENTITY"

$$X = \begin{array}{c c} & \begin{array}{cc}\langle z_+| & \langle z_-|\end{array} \\ & \begin{pmatrix} 0 & 1 \\ 1 & 0\end{pmatrix} \begin{array}{c}|z_+\rangle \\ |z_-\rangle\end{array}\end{array}$$

$$Y = \begin{array}{c c} & \begin{array}{cc}\langle z_+| & \langle z_-|\end{array} \\ & \begin{pmatrix} 0 & -i \\ i & 0\end{pmatrix} \begin{array}{c}|z_+\rangle \\ |z_-\rangle\end{array}\end{array}$$

"IDENTITY WITH A MINUS"

$$Z = \begin{array}{c c} & \begin{array}{cc}\langle z_+| & \langle z_-|\end{array} \\ & \begin{pmatrix} 1 & 0 \\ 0 & -1\end{pmatrix} \begin{array}{c}|z_+\rangle \\ |z_-\rangle\end{array}\end{array}$$

- $X = |z_-\rangle\langle z_+| + |z_+\rangle\langle z_-|$
- $Y = -i|z_+\rangle\langle z_-| + i|z_-\rangle\langle z_+|$
- $Z = |z_+\rangle\langle z_+| - |z_-\rangle\langle z_-|$

- $S_x = \frac{\hbar}{2}|z_-\rangle\langle z_+| + \frac{\hbar}{2}|z_+\rangle\langle z_-| = \frac{\hbar}{2}X$
- $S_y = -\frac{\hbar}{2}i|z_+\rangle\langle z_-| + \frac{\hbar}{2}i|z_-\rangle\langle z_+| = \frac{\hbar}{2}Y$
- $S_z = \frac{\hbar}{2}|z_+\rangle\langle z_+| - \frac{\hbar}{2}|z_-\rangle\langle z_-| = \frac{\hbar}{2}Z$

• EXPECTED VALUE AND VARIANCE

- $\langle A\rangle = \langle\psi|A|\psi\rangle$; $\langle A\rangle = \psi^\dagger A\psi$ $\quad \langle A\rangle = Tr\rho A$
- $\langle A^k\rangle = \langle\psi|A^k|\psi\rangle$
- $\langle\Delta A^2\rangle = \langle A^2\rangle - \langle A\rangle^2$
- $\Delta S_x = \sqrt{\langle S_x^2\rangle - \langle S_x\rangle^2}$ ⟵ STANDARD DEVIATION
- $\Delta S_x^2 = \langle S_x^2\rangle - \langle S_x\rangle^2$ ⟵ VARIANCE

- COMMUTATOR: $[A,B] = AB - BA \longrightarrow$ IF $A = A^\dagger$ AND $B = B^\dagger \Rightarrow [A,B]^\dagger = -[A,B]$ (ANTIHERMITIAN)
- ANTICOMMUTATOR: $\{A,B\} = AB + BA \longrightarrow$ IF $A = A^\dagger$ AND $B = B^\dagger \Rightarrow \{A,B\}^\dagger = \{A,B\}$ (HERMITIAN)
- $AB = \frac{1}{2}[A,B] + \frac{1}{2}\{A,B\}$

COMPLEX NUMBERS

- $i = \sqrt{-1}$; $i^2 = -1$
- $e^{i\pi} = -1$
- $\alpha\alpha^* = |\alpha|^2$
- $\alpha + \alpha^* = 2RE(\alpha)$
- $|3-5i|^2 = 3^2 + 5^2$
- $e^{ix} = \cos(x) + i\sin(x)$
- $|-5i| = |5i| = 5$

(NOTE: $\langle\psi|[A,B]|\psi\rangle$ IS WRITTEN AS $\langle[A,B]\rangle$)

• UNCERTAINTY RELATION. $|\langle\psi|FG|\psi\rangle|^2 \geq \frac{1}{4}|\langle\psi|[F,G]|\psi\rangle|^2$ $\qquad \Delta A\Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$

• UNITARY EVOLUTION

OPERATOR OF TIME EVOLUTION OF INFORMATIONALLY ISOLATED SYSTEMS OVER THE INTERVAL t_1 TO t_2

- $|\psi(t_2)\rangle = U(t_2,t_1)|\psi(t_1)\rangle$

$U = \begin{pmatrix} e^{-i\omega_0 t} & 0 \\ 0 & e^{-i\omega_1 t}\end{pmatrix}$

- $|\psi'\rangle = U|\psi\rangle$; $|\phi'\rangle = U|\phi\rangle \longrightarrow \langle\psi'|\phi'\rangle = \langle\psi|\overbrace{U^\dagger U}^{?}|\phi\rangle = \langle\psi|\phi\rangle$
- $U(t_3,t_1) = U(t_3,t_2)U(t_2,t_1)$

$H: \begin{pmatrix}\omega_0 & 0\\0 & \omega_1\end{pmatrix}$ $\quad \omega_0 = \frac{H}{\hbar}$ $\quad \omega_1 = \frac{H}{\hbar}$

- $|\psi(t)\rangle = \alpha_0 e^{-i\omega_0 t}|E_0\rangle + \alpha_1 e^{-i\omega_1 t}|E_1\rangle$
- $|\psi(t)\rangle = U(t)|\psi(0)\rangle$

- $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$
- $Tr[|a\rangle\langle b|] = \langle a|b\rangle$

6. ENTANGLEMENT $\langle a_1,b_1|a_2,b_2\rangle = \langle a_1|a_2\rangle\langle b_1|b_2\rangle$

JOINT PROBABILITY: $\begin{cases} \bullet\ P(a,b) = |\langle a,b|\psi^{(AB)}\rangle|^2 \\ \bullet\ P(a,b) = |\langle a|\phi_b\rangle|^2 \end{cases} \longrightarrow$

DISTRIBUTION OVER a AND b INDIVIDUALLY

$P(a) = \sum_b P(a,b)$ $\quad = \langle\phi_a|\phi_a\rangle$

PROBABILITY OF MEASURING a REGARDLESS OF THE RESULT ON b. I FIX a AND CHANGE b

$P(b) = \sum_a P(a,b)$ $\quad = \sum_a |\langle a|\phi_b\rangle|^2 = \sum_a \langle\phi_b|a\rangle\langle a|\phi_b\rangle = \langle\phi_b|\phi_b\rangle$

PROBABILITY OF MEASURING b REGARDLESS OF THE RESULT ON a. I FIX b AND CHANGE a

REMEMBER: $\sum_n |n\rangle\langle n| = \mathbb{1}$

$|\psi^{(AB)}\rangle = \sum_{a,b} x_{a,b}|a\rangle\otimes|b\rangle = \sum_b |\phi_b\rangle\otimes|b\rangle$

$\sum_a x_{a,b}|a\rangle$ IS A NON-NORMALIZED VECTOR IN $\mathcal{H}^{(A)}$

• DENSITY MATRIX

$\rho = \begin{pmatrix} a & b \\ c & d\end{pmatrix}$ $\quad \boxed{a + d = 1}$

$\rho\begin{pmatrix} a & 0 \\ 0 & b\end{pmatrix} \longrightarrow$ MIXED STATE $\quad Tr(\rho^2) < 1$

- PURE STATE: $Tr(\rho^2) = 1$
- MIXED STATE: $Tr(\rho^2) \neq 1$

$\rho = |\psi\rangle\langle\psi|$

$\langle S_x\rangle = Tr(\rho S_x)$

$\langle A\rangle = Tr(\rho A)$

BLOCK VECTOR: FIND ρ OF A STATE ($\rho = |\psi\rangle\langle\psi|$) AND THEN:

$\rho = \begin{pmatrix} a & b \\ c & d\end{pmatrix} \longrightarrow \begin{cases} x = 2RE(b) \\ y = 2I(c) \\ z = a - d \end{cases}$

OPERATORS AND EIGEN

- $\hat{A}|\psi\rangle = |\psi'\rangle$

USUALLY $\hat{A}|\psi\rangle = \lambda|\psi\rangle$ (SCALAR)

- $|\psi\rangle \longrightarrow$ EIGENSTATE/EIGENVECTOR OF $\hat{A}$
- $\lambda \longrightarrow$ EIGENVALUE OF $\hat{A}$

$\hat{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0\end{pmatrix}$ $\quad |\psi\rangle = \begin{pmatrix} c_1 \\ c_2\end{pmatrix}$ FIND THE EIGENVALUES AND EIGENVECTOR

SAME OF EIGENSTATE

$\hat{A}|\psi\rangle = \lambda|\psi\rangle$

- FIND EIGENVALUES

$\det(\hat{A} - \lambda I) = \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda\end{pmatrix} = -\lambda^2 + 1 \Rightarrow \lambda^2 = -1 \Rightarrow \lambda_1 = i;\ \lambda_2 = -i$

- FIND EIGENVECTOR FOR $\lambda_1 = i$

$\hat{A}|\psi\rangle = \lambda_1|\psi\rangle;\ \begin{pmatrix} 0 & -1 \\ 1 & 0\end{pmatrix}\begin{pmatrix} c_1 \\ c_2\end{pmatrix} = i\begin{pmatrix} c_1 \\ c_2\end{pmatrix};\ \begin{cases} -c_2 = ic_1 \\ c_1 = ic_2\end{cases} \begin{cases} ic_1 + c_2 = 0 \\ c_1 - ic_2 = 0;\ ic_1 + c_2 = 0\end{cases} \longrightarrow \begin{cases} c_1 = i \\ c_2 = 1\end{cases} \longrightarrow \begin{pmatrix} i \\ 1\end{pmatrix} \xrightarrow{\text{NORMALIZE}} \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1\end{pmatrix}$

EIGENVECTOR

ANY MULTIPLE OF THE EIGENVECTOR IS A VALID EIGENVECTOR TOO

- FIND EIGENVECTOR FOR $\lambda_2 = -i$

$\hat{A}|\psi\rangle = \lambda_2|\psi\rangle;\ \begin{pmatrix} 0 & -1 \\ 1 & 0\end{pmatrix}\begin{pmatrix} c_1 \\ c_2\end{pmatrix} = -i\begin{pmatrix} c_1 \\ c_2\end{pmatrix};\ \begin{cases} -c_2 = -ic_1 \\ c_1 = -ic_2\end{cases} \begin{cases} -ic_1 + c_2 = 0 \\ c_1 + ic_2 = 0\end{cases};\begin{cases} c_1 + ic_2 = 0 \\ c_1 + ic_2 = 0\end{cases} \longrightarrow \begin{cases} c_1 = 1 \\ c_2 = i\end{cases} \longrightarrow \begin{pmatrix} 1 \\ i\end{pmatrix} \xrightarrow{\text{NORMALIZE}} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i\end{pmatrix}$

EIGENVECTOR

The Hamiltonian operator is a fundamental concept in quantum mechanics that describes the total energy of a system. In mathematical terms, it is a linear operator that acts on the wave function of the system and yields the total energy as an eigenvalue. The Hamiltonian operator can be used to derive the time evolution of a quantum system and to predict the outcomes of measurements.

Hilbert space is a fundamental concept in quantum mechanics and has several important properties. Here are some of the key properties of Hilbert space:

1. Inner product: [illegible]
2. Completeness: [illegible]
3. Orthonormal basis: [illegible]
4. Continuous linear transformations: [illegible]
5. Uncertainty principle: [illegible]

In summary, Hilbert space is a rich mathematical structure with several important properties that underlie many of the key concepts and results in quantum mechanics.

Information

Information is the ability to **distinguish reliably** between possible alternatives (physical signals).

Code is a **one to one function** that relates the message value to the signal (in the case of lamps to represent British attacks in American revolution).

A message can have more than one representation, and information can be transferred from one physical representation to another.

Bit is a generic term we will use for a physical system with two possible distinguishable states (2 is the minimum to perform a distinction). **Byte** is composed by 8 bits.

If M is the number of possible values of a message, we define **entropy** of the message as:

$$H = \log_2(M)$$

It measures the **information necessary** (minimum bits necessary) to encode the message. When it results in a not integer float, we round to the upper number.

If we have two independent messages A and B:

$$H(AB) = H(A) + H(B)$$

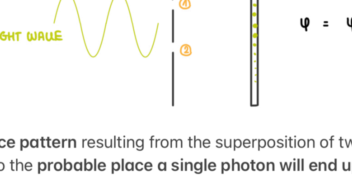
Noise affects our capability to distinguish between reliable and non reliable information. It's defined as anything that **prevents a signal to be ready unambiguously**.

$$H - H' = \log\left(\frac{m}{m'}\right)$$

POSSIBLE MESSAGE VALUES

MESSAGE VALUES THAT CAN BE READ

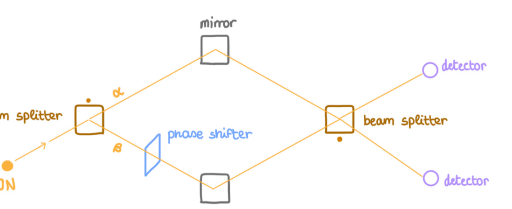
Young's experiment



$$\varphi = \varphi_1 + \varphi_2$$

the **interference pattern** resulting from the superposition of two light waves corresponds to the **probable place a single photon will end up** when shoot through one of the two slits (almost like it's interfering with itself)

Interferometer



INITIAL STATE $|\Psi\rangle = \begin{pmatrix}\alpha \\ \beta\end{pmatrix}$ probability amplitudes $\alpha\begin{pmatrix}1 \\ 0\end{pmatrix} + \beta\begin{pmatrix}0 \\ 1\end{pmatrix}$

FINAL STATE $|\Psi'\rangle = I\,|\Psi\rangle = \begin{pmatrix}\alpha' \\ \beta'\end{pmatrix}$

interferometer matrix

$$I = BS_2\, PS(\varphi)\, BS_1$$

$$\begin{pmatrix}\alpha' \\ \beta'\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}-1 & 1 \\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 \\ 0 & e^{i\varphi}\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1 \\ 1 & -1\end{bmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix}$$

Kets and Bras

In quantum mechanics, **particles can exist in a superposition of different states**, which means that they can have multiple values of observable quantities such as momentum, energy, and position simultaneously. The **wave function** of a particle (which describes the state of the particle at any given time) can be written as a **sum of states** with different momenta and energies, each weighted by a corresponding **probability amplitude**.

$$|\psi\rangle = a|a\rangle + b|b\rangle$$

When we measure the momentum or energy of the particle, the wave function collapses to one of the possible states (wave/particle duality), with a **probability** given by the **square of its amplitude**.

$$P(a) = |a|^2$$

In summary, a probability amplitude tells us how much of each state is present in the overall wave function of a particle.

In braket notation, a **quantum state** is represented by the **ket**, a particular **vector** in a complex vector space called a Hilbert space. The elements of the vector correspond to the probability amplitudes of the different possible outcomes of a measurement of the system.

$$|\Psi\rangle$$

The braket notation also includes a dual space called the **bra**, a linear functional that maps a ket to a complex number.

$$\langle\Psi|$$

Together, the bra and the ket form a **bra-ket**, or **inner product**, which is a complex number that represents the **probability amplitude of the transition from the state represented by the ket to the state represented by the bra**. In other words, it gives the overlap between the two states.

$$\langle u|\psi\rangle$$

HERMITIAN CONJUGATE

$$u^\dagger\psi = [u_1^*\ u_2^*\ u_3^* \dots]\begin{bmatrix}\psi_1 \\ \psi_2 \\ \psi_3 \\ \dots\end{bmatrix}$$

if we want to calculate the **probability** that a measurement of a certain property results in the **outcome u**:

$$P_u = |\langle u|\psi\rangle|^2$$

PROJECTION

since any measurement is like **projecting the system into a different reference**.

the general **state of an atom** is represented by:

$$|\psi\rangle = \alpha_0|E_0\rangle + \alpha_1|E_1\rangle$$

GROUND EXCITED

$|\alpha_0|^2$ PROBABILITY OF STATE $|E_0\rangle$

SCHWARTZ INEQUALITY: $\langle a|a\rangle\langle b|b\rangle \geq |\langle a|b\rangle|^2$

States

Qubit: two level quantum system (represented in a 2 dimensional Hilbert Space)
Hilbert Space: complex vector space spanned by orthonormal basis (set of normal vectors orthogonal to each other).
Ket: vector representation of a quantum state (normalized).
Operators: functions on the Hilbert space that map vectors to vectors linearly.

$$|\psi\rangle = \sum_i c_i|i\rangle$$

UNITARY TIME EVOLUTION

Schrödinger's equation: controls the time evolution of a quantum state.

$$i\hbar\frac{d}{dt}|\Psi(t)\rangle = H|\Psi(t)\rangle$$

H → Hamiltonian operator: returns total energy of the system

i → imaginary unit: $\sqrt{-1}$

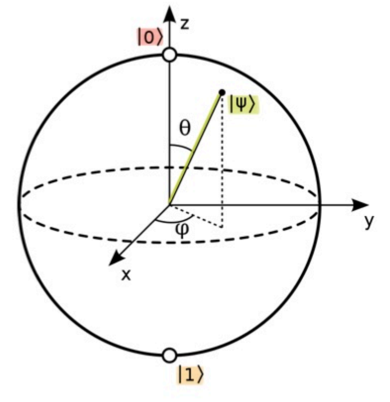
$\hbar$ → reduced Planck's constant: $h/2\pi$ where $h = 6.626 \dots 10^{-34}$

Qubit

Q

Qubit Hilbert space

TWO DIMENSIONAL → BASIS $\{|0\rangle, |1\rangle\}$



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

different kinds of states in qubit basis $\{|0\rangle, |1\rangle\}$

SINGLE PURE STATE: linear combination of basis states $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

MULTIPLE PURE STATES: more than one pure states considered together

1. separable states
2. entangled states

MIXED STATES: incomplete knowledge about single states

Operators

→ OPERATORS ON Q

$\langle 1|0\rangle = 0$ $\quad\langle 1|1\rangle = 1$

since basis states are orthogonal

PROJECTION	Π_K	→ $\|K\rangle\langle K\|$		picks out part parallel to $\|K\rangle$
IDENTITY	I	→ $\|0\rangle\langle 0\| + \|1\rangle\langle 1\|$	$\mathbb{1} = \begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix}$	returns state itself
BIT FLIP	X	→ $\|0\rangle\langle 1\| + \|1\rangle\langle 0\|$	$\sigma_x = \begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix}$	swaps amplitudes
PHASE SHIFT	Y	→ $-i\|0\rangle\langle 1\| + i\|1\rangle\langle 0\|$	$\sigma_y = \begin{pmatrix}0 & -i \\ i & 0\end{pmatrix}$	rotates around y by 90°
PHASE FLIP	Z	→ $\|0\rangle\langle 0\| - \|1\rangle\langle 1\|$	$\sigma_z = \begin{pmatrix}0 & 1 \\ -1 & 0\end{pmatrix}$	inverts sign of the amplitude

PROPERTIES:

$$\mathrm{Tr}\,\sigma = 0$$

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = -i\,\sigma_x\sigma_y\sigma_z = \mathbb{1}$$

PROPERTIES OF OPERATORS

linearity: $G\left(\sum a_n|\varphi_n\rangle\right) = \sum a_n G|\varphi_n\rangle$

Kernel: $\ker(A) = \{|\psi\rangle : A|\psi\rangle = 0\}$

commutator: $[A,B] = AB - BA = -[B,A]$ with $[A,A] = 0$

when a basis is specified, an operator can be represented by a **matrix**:

$G = \{G|1\rangle, G|2\rangle, \dots G|n\rangle\}$ $\quad G_{mn} = \langle m|G|n\rangle$

OPERATORS ARE USED TO FIND PROBABILITY OF THE POSSIBLE VALUES OF THE OBSERVABLES

$$\begin{array}{c|ccc} & |1\rangle & \cdots & |n\rangle \\ \hline \langle 1| & G_{11} & \dots & G_{1d} \\ \dots & \dots & & \dots \\ \langle n| & G_{d1} & \dots & G_{dd} \end{array}$$

EXPECTATION $\langle A\rangle \rightsquigarrow \langle\psi|A|\psi\rangle$

VARIANCE $\langle\Delta A^2\rangle \rightsquigarrow \langle A^2\rangle - \langle A\rangle^2$

UNCERTAINTY $\Delta A \rightsquigarrow \sqrt{\langle\Delta A^2\rangle}$

UNCERTAINTY RELATION

$$\Delta A\,\Delta B \geq \frac{1}{2}|\langle[A,B]\rangle| \rightsquigarrow \langle\Delta A^2\rangle\langle\Delta B^2\rangle \geq \frac{1}{4}|\langle\psi|[A,B]|\psi\rangle|^2$$

when two observable commute ($[A,B] = 0$) they can be measured with arbitrary precision

DELTA FUNCTION

KRONECKER DELTA SYMBOL $\delta_{mn} \rightsquigarrow 1$ when $m = n$, 0 otherwise $\rightsquigarrow \sum_n \delta_{mn}a_n = a_m$

DIRAC DELTA FUNCTION $\delta(x, x_0) \rightsquigarrow 0$ when $x \neq x_0$, integrates to 1 $\rightsquigarrow \int_x \delta(x, x_0)\,f(x)\,dx = f(x_0)$

PROJECTIONS WITH δ

$|\Psi\rangle = \sum a_n|n\rangle$ with $n \in \{0,1\}$ $\qquad |\Psi\rangle = \int_x \Psi(x)|x\rangle\,dx$ with $x \in [\text{position}]$

$$\Pi_0|\Psi\rangle = \sum_n a_n|0\rangle\langle 0|n\rangle = \sum_n \delta_{0n}a_n|n\rangle = a_0|0\rangle$$

$$\Pi_{x_0}|\Psi\rangle = \int_x \Psi(x)\,|x_0\rangle\langle x_0|x\rangle\,dx = \int_x \delta(x, x_0)\,\Psi(x)\,|x_0\rangle\,dx = \Psi(x_0)|x_0\rangle$$

↓

$\delta_{mn} = \langle m|n\rangle$ $\qquad \delta(x, x_0) = \langle x_0|x\rangle$

SPIN

intrinsic angular momentum (NO ACTUAL ROTATION) of particles

quantized ⇝ can only have discrete values ⇝ MULTIPLES OF $\hbar$

SPIN 1/2 PARTICLES ⇝ can have values $\pm\left(\frac{1}{2}\right)\hbar$

we can measure each component with the corresponding PAULI OPERATOR:

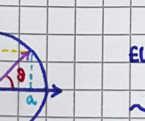
$S_x = \frac{\hbar}{2}\sigma_x$ $\qquad S_y = \frac{\hbar}{2}\sigma_y$ $\qquad S_z = \frac{\hbar}{2}\sigma_x$

RECAP complex numbers

$z = a + ib$ $\qquad \bar{z} = z^* = a - ib$

with $\mathrm{Re}(z) = a$ $\quad \mathrm{Im}(z) = b$

magnitude $|z| = \sqrt{\mathrm{Re}(z)^2 + \mathrm{Im}(z)^2} = \sqrt{a^2 + b^2}$



EULER'S IDENTITY: $e^{i\varphi} = \cos\varphi + i\sin\varphi$

$\rightsquigarrow z = \rho e^{i\vartheta} = \rho(\cos\vartheta + i\sin\vartheta)$

with $a = \rho\cos\vartheta$ and $b = \rho\sin\vartheta$

OBSERVABLE O ⇝ eigenvalues: POSSIBLE OUTCOMES

O_{nn} $\qquad$ eigenvectors: POSSIBLE STATES OF THE SYS

with n → number of basis vectors $\{|n\rangle\}$

$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix}\alpha \\ \beta\end{pmatrix}$

$O|\Psi\rangle = [\quad]\begin{pmatrix}\alpha \\ \beta\end{pmatrix}$

$\langle 0|$, $\langle 1|$, $|0\rangle$ $|1\rangle$ ----> eigenvectors · eigenvalues

PRODUCTS

$|\varphi\rangle \quad |\psi\rangle$

$\begin{bmatrix}1 \\ 2\end{bmatrix} \quad \begin{bmatrix}1 \\ 2\end{bmatrix}$

outer $\qquad$ inner

$|\varphi\rangle\langle\psi| \qquad \langle\varphi|\psi\rangle$

PRODUCT

SUM

CHANGING BASIS

$|x_+\rangle = 1/\sqrt{2}\,(|y_+\rangle - i|y_-\rangle) = 1/\sqrt{2}\,(|z_+\rangle + |z_-\rangle)$

$|x_-\rangle = 1/\sqrt{2}\,(|y_+\rangle + i|y_-\rangle) = 1/\sqrt{2}\,(|z_+\rangle - |z_-\rangle)$

$|y_+\rangle = 1/\sqrt{2}\,(|x_+\rangle - i|x_-\rangle) = 1/\sqrt{2}\,(|z_+\rangle + i|z_-\rangle)$

$|y_-\rangle = 1/\sqrt{2}\,(|x_+\rangle + i|x_-\rangle) = 1/\sqrt{2}\,(|z_+\rangle - i|z_-\rangle)$

$|z_+\rangle = 1/\sqrt{2}\,(|x_+\rangle + |x_-\rangle) = 1/\sqrt{2}\,(|y_+\rangle + i|y_-\rangle)$

$|z_-\rangle = 1/\sqrt{2}\,(|x_+\rangle - |x_-\rangle) = 1/\sqrt{2}\,(|y_+\rangle - i|y_-\rangle)$

$$\mathbb{1} = \int_{-\infty}^{+\infty}|x\rangle\langle x|\,dx$$

$$\langle x|\phi\rangle = \langle x|A|\psi\rangle = \langle x|\int dx'\,A|x'\rangle\langle x'|\psi\rangle = \int dx'\,\underbrace{\langle x|A|x'\rangle}_{A(x,x')}\underbrace{\langle x'|\psi\rangle}_{\psi(x')}$$

$$\boxed{\phi(x) = \int dx'\,A(x,x')\,\psi(x')}$$

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}e^{-ipx/\hbar}\psi(x)\,dx \qquad p = \hbar k$$

$$= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}e^{-ikx}\psi(x)\,dx$$

$$\tilde{F}(k) = \int_{-\infty}^{+\infty}e^{-ikx}f(x)\,dx \qquad \text{FOURIER TRANSFORM}$$

What about operators?

	$\{\|n\rangle\}$	$\{\|x\rangle\}$
$\|\psi\rangle$	$\psi_n = \langle n\|\psi\rangle$	$\psi(x) = \langle x\|\psi\rangle$
A	$A_{mn} = \langle m\|A\|n\rangle$	$A(x,x') = \langle x\|A\|x'\rangle$
$\|\phi\rangle = A\|\psi\rangle$	$\phi_m = \sum_n A_{mn}\psi_n$	$\phi(x) = \int_{-\infty}^{+\infty}dx'\,A(x,x')\psi(x')$

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DISCRETE STATES

$|\psi\rangle = \sum a_k|k\rangle$

PROBABILITY VALUE $|a_k|^2$

CONTINUOUS STATES

$|\Psi\rangle = \int_{-\infty}^{\infty}\Psi(x)|x\rangle\,dx$

PROBABILITY DENSITY $|\Psi(x)|^2$

⇝ KINDS OF WAVE FUNCTIONS:

1. GAUSSIAN 2. LAURENTIAN $(?) = \left[1/\pi(1+x^2)\right]$

$\Psi(x)$ COMPLEX VALUED ⇝ $\mathrm{Re}(\Psi(x)) + \mathrm{Im}(\Psi(x))$

$\Psi(x) = |\Psi(x)|\,e^{i\varphi}$ PHASE DOES NOT DEPEND ON THE POSITION

$= |\Psi(x)|\,(\cos\varphi + i\sin\varphi)$ PHASE SHIFT OF π/2

3. DELTA FUNCTION $\delta(x - x')$ $\quad |\Psi\rangle = \int_{-\infty}^{\infty}\Psi(x)|x\rangle\,dx$

$\langle x|x'\rangle = \delta(x - x')$ ⇝ ORTHONORMALITY

$1 = \int_{-\infty}^{\infty}dx\,|x\rangle\langle x|$ ⇝ COMPLETENESS

ALSO DISCRETE $\sum_k |k\rangle\langle k| = 1$

PROPERTIES: $x \neq x' \rightarrow \delta = 0$; integrates to 1